TRINETRA* GUARD — Complete Master Brief v4

BGI Hackathon 2026 · Problem Statement IT4P2 · Emergency Health Monitoring Wearable

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SECTION 1 — PROJECT IDENTITY

- **Competition:** BGI Hackathon 2026, Indore, 15–16 May 2026
- **Problem Statement:** IT4P2 — Emergency Health Monitoring Wearable
- **Category:** Hardware (Primary) with Software Integration
- **Previous project:** TRINETRA ring — too small, no GSM, no ECG, no fall detection. Pivoted to screenless magnetic strap band.
- **Regulatory framing:** "Health monitoring and emergency alert wearable." NEVER "medical device." We monitor and alert. We do not diagnose or treat.

SECTION 2 — WHAT IT IS

A screenless emergency health monitoring wristband with a detachable magnetic electronics puck (65mm × 28mm × 15mm — updated Session 4). No display — feedback via RGB LED + haptic vibration. Phone app shows all data. When a critical event occurs, automatic WhatsApp alert fires to caregiver with GPS location. If patient is unconscious and cannot cancel — the dead man's switch fires anyway.

SECTION 3 — PRIORITY STACK (MOST IMPORTANT SECTION)

TIER 1 — NON-NEGOTIABLE. Must work perfectly. No exceptions.

1. Heart rate + SpO2 live on app via BLE
2. Fall detection algorithm firing correctly
3. Dead man's switch countdown visible on app
4. Twilio WhatsApp alert firing on a real phone in the room with GPS location

TIER 2 — These marks win for my team

5. ECG waveform visible on app (even slightly noisy — judges aren't cardiologists)
6. Physical puck on wrist, magnetically detaching and reattaching cleanly
7. Temperature + heat stress index on app

TIER 3 — Add only when Tier 1 and 2 are solid.

8. Chest strap swap demonstration
9. TEG voltage on dashboard
10. Adaptive baseline, sleep apnea screening, medication reminders

The trap: Building everything to 60% completion. For tier 3 the Tier 1 should be at 100%, Tier 2 at 90%,

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SECTION 4 — FORM FACTOR

Primary: Wrist Band (BUILD THIS — DEMO THIS)

- 22mm silicone or dark elastic band
- Puck snaps via 4× N52 neodymium magnets (6mm×3mm) at corners
- 2× brass alignment pins prevent rotation
- 2× pogo pins auto-complete ECG circuit on attachment
- 3 wire grooves in inner band for electrode wiring, sealed with silicone

Secondary: Chest Strap (Day 16 ONLY if wrist band works Day 15)

Same puck, 5cm elastic chest strap, identical magnet dock. If Day 15 is rough — chest strap never exists

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SECTION 5 — SENSOR STACK

#	Sensor	Parameters	Interface	Address	Cost (INR) (tentative..subject to increase if willing to pay)
1	MAX30102	HR, SpO2, HRV	I2C	0x57	₹150–250
2	MPU6050 GY-521	Fall, Activity, Sedentary	I2C	0x68	₹80–130
3	MLX90614	Skin + ambient temp, Heat stress	I2C	0x5A	₹280–420
4	AD8232	Single-lead ECG, QRS, AF flag	Analog	—	₹250–400
5	BMP280	Pressure/altitude, fall confirm	I2C	0x76	₹60–100
6	DS3231 RTC	Timekeeping, medication	I2C	—	₹80–120
7	Micro-TEG + BQ25504	Body heat harvesting	Analog	—	₹200–350

All I2C on shared bus. AD8232 analog. Zero address conflicts.

CUT — Final

Cut	Reason
GSR electrodes	ESP32 ADC noise = constant false positives
SIM800L GSM	2A peak brownout. Phone is gateway.
ADXL362 2nd IMU	Redundant with MPU6050
40mm TEG	Larger than puck
Abdomen band	No time for 2nd hardware variant

SECTION 6 — PHYSICAL STRUCTURE (UPDATED v4)

Puck Dimensions: 65mm × 28mm × 15mm

CHANGE FROM v3: Dimensions updated from 55×30×13mm to 65×28×15mm.

- Longer + narrower = WHOOP-inspired silhouette. Reads as a wristband, not a box.
- +2mm height accommodates AD8232 module stacking without bare-IC SMD soldering.
- Same internal volume. Totally different visual language on the wrist.

TOP FACE:

- Stainless steel ECG finger pad (20×20mm) — center, flush-mounted
- RGB LED SMD 5050 — top-left, recessed 1mm into channel (only glow visible, not LED itself)
- SOS button 3mm tactile — top-right, flush
- All top edges: 2mm chamfer (critical for non-boxy appearance)
- USB-C port — recessed on right side edge

MIDDLE LAYER (perfboard 5×7cm):

- ESP32-S3 Mini (25×18mm)
- AD8232 ECG module
- DS3231 RTC
- BQ25504 harvester

- MT3608 boost + TP4056 charging
- 2N2222 transistors

BATTERY: LiPo 1000mAh flat (34×22×5mm) between layers, center for weight balance

BOTTOM FACE (skin side):

- MAX30102 center aperture
- MLX90614 1cm beside MAX30102
- Micro-TEG (15×15mm) thermal paste
- 4× N52 magnets at corners (Araldite-potted, 24hr cure)
- 2× pogo pins for ECG circuit
- Bottom face: NO chamfer (flat against wrist)

Band:

- Material: Dark grey or black elastic fitness band (from Decathlon or sports shop)
- NOT generic silicone strap — elastic reads as designed, not assembled
- Groove 1: Wrist pad → wire → pogo → AD8232 IN+
- Groove 2: RL drive pad → wire → pogo → AD8232 RL
- Groove 3: Reserved v2
- Seal with medical silicone

Magnet System:

- 4× N52, 6mm×3mm, Araldite-potted
- 2× 2mm brass alignment pins
- ~4.8kg hold force total
- Sensors at center, magnets at corners — test interference Day 14

Step 1: Cardboard Mockup (TODAY — 30 minutes)

Before any CAD or printing:

- Cut cardboard to 65mm × 28mm × 15mm

- Stack 3–4 layers to get height
- Tape together
- Put on your wrist. Walk around for 10 minutes.
- If it feels too long → go to 60mm. If too wide → go to 26mm.
- **Your wrist decides the final dimensions, not the spec sheet.**
- Take a photo and confirm dimensions before going to print shop.

Step 2: Find 3D Print Shop (TODAY)

- Search Google Maps: "3D printing service Indore"
- Check near: Vijay Nagar, Palasia, IET/DAVV campus area
- **Walk in. Don't call. Show up with paper sketch.**
- Tell them: hackathon deadline, need it in 48 hours
- Cost: ₹200–500. Pay whatever they ask.
- No CAD skill needed on your team — the shop models it from your sketch.

Step 3: Paper Sketch to Hand the Print Shop

Draw this on paper. Photo it. Confirm with Claude before walking in.

The box: 65mm long × 28mm wide × 15mm tall

Top face cutouts:

- Center: 22×22mm square recess, 1mm deep (ECG steel pad)
- Top-left: 3mm circle hole, recessed 1mm (LED channel)
- Top-right: 4mm circle hole (SOS button)
- All top edges: 2mm chamfer/bevel

Side edge cutout:

- Right side, 10mm from top-right corner: USB-C slot, 10mm wide × 4mm tall

Bottom face cutouts:

- Center: 10mm circle hole (MAX30102)
- 15mm left of center: 8mm circle hole (MLX90614)

- Four corners: 7mm circle holes, 3mm deep (magnets)
- Two 2mm holes near center edge (pogo pins)
- Bottom face: flat, no chamfer

Step 4: Finish Protocol (What Makes It Look Like a Product)

Print in matte black PLA. Then:

Day 1 evening (after collecting print):

- Sand with 220 grit. Then 400 grit. 20 minutes total.

Day 2 morning:

- One coat grey automotive primer spray (Motip or any auto shop brand)
- Let dry 4 hours. Do not touch.

Day 2 evening:

- Two coats matte black spray (Rust-Oleum 2X Matte or equivalent)
- 30 minutes between coats

Cost: ₹150–200 for spray cans. Available at any hardware shop near Khajrana or Palasia.
Result: Indistinguishable from injection-moulded plastic at 1 metre distance.

Why This Works Aesthetically

The four things that make a puck look premium — not size:

1. Matte surface, zero gloss
2. Chamfered top edges (looks designed, not printed)
3. Flush stainless steel ECG pad (signals "serious hardware" under light)
4. Band color intentionally matched — matte black puck + dark grey elastic band

The LED recessed into a channel: when it pulses blue in the demo, judges feel it before they understand it.

SECTION 7 — COMPLETE BOM

Core Electronics

Component	Spec	Cost (INR)	Source
ESP32-S3 Mini	25×18mm, BLE 5.0	₹400–600	Robu.in
ESP32 DevKit v1	Development + breadboard fallback	₹350–450	Robu.in
MAX30102	I2C HR+SpO2	₹150–250	Robu.in
MPU6050 GY-521	I2C Fall+Activity	₹80–130	Robu.in
MLX90614	I2C ±0.5°C contactless	₹280–420	Robu.in
AD8232 module	Analog ECG	₹250–400	Amazon/Robu.in
BMP280	I2C pressure	₹60–100	Robu.in
DS3231 RTC	I2C battery-backed	₹80–120	Robu.in
LiPo 3.7V 1000mAh flat	34×22×5mm	₹200–350	Amazon India
TP4056 USB-C	LiPo charging	₹30–60	Robu.in
MT3608 boost	3.7V→5V — CRITICAL, NOT OPTIONAL	₹30–60	Robu.in
Micro-TEG 15mm×15mm	Bi-Te	₹80–150	LCSC/AliExpress
BQ25504	MPPT harvester	₹120–200	LCSC/Robu.in
1F 5.5V supercap	TEG buffer	₹60–100	Amazon

Feedback + Interface

Component	Cost (INR)
RGB LED SMD 5050 common cathode	₹10–30

Component	Cost (INR)
ERM coin vibration motor 10mm	₹50–80
Tactile button 6mm flush	₹5–15
Stainless steel pads ×3 (20×20mm, 304 SS)	₹80–150
Pogo pins ×4 spring-loaded 2mm	₹60–100

Mechanical

Component	Cost (INR)
N52 magnets 6×3mm (20 pcs)	₹150–250
Brass alignment pins 2mm ×4	₹20–50
Dark elastic band 28mm wide (Decathlon)	₹100–200
Elastic chest strap 5cm (Day 16 only)	₹100–200
3D print (outsourced to local shop)	₹200–500
Araldite epoxy	₹60–100
Thermal paste	₹80–120
Perfboard 5×7cm ×3	₹90–180
Silicone sealant	₹60–100
Grey primer spray	₹80–120
Matte black spray (Rust-Oleum 2X)	₹150–200
Sandpaper 220+400 grit	₹30–50

Passives + Wiring

- 4.7kΩ resistors ×10: ₹10–20
- 2N2222 NPN ×5: ₹10–20
- 100µF caps ×4: ₹20–40
- 28–30 AWG enamelled wire: ₹30–50

- Heat shrink tubing: ₹30–50
- Jumper wires: ₹50–80

TOTAL: ₹2,800–4,200

SECTION 8 — FEATURES

Feature	Tier	Sensor	Alert	Demo-able?
Heart rate	T1	MAX30102	<45 or >130 bpm	Yes — live
SpO2	T1	MAX30102	<90% critical	Yes — live
Fall detection	T1	MPU6050 + BMP280	Accel >2.5g + stillness 800ms	Yes — drop
Dead man's switch	T1	All	No cancel in 30s	Yes — countdown
WhatsApp alert + GPS	T1	Phone GPS + Twilio	On Tier 3 event	Yes — judge's phone
ECG + AF flag	T2	AD8232	Irregular RR	Yes — waveform
Magnetic puck demo	T2	Physical	—	Yes — detach/ reattach
Skin temperature	T2	MLX90614	>38°C / <35°C	Yes — live
Heat stress index	T2	MLX90614 dual	Skin < ambient by >2°C	Yes
HRV / stress	T3	MAX30102	Sustained low HRV	Yes — app graph
Manual SOS	T3	Button 3s	Immediate	Yes
Sleep apnea screening	T3	MAX30102	3+ dips <90%	Yes — morning report
Medication reminder	T3	DS3231 + vibration	Timed haptic	Yes
TEG harvesting	T3	Micro-TEG	Live µW display	Yes — dashboard
Context mode switch	T3	Button 5s	LED color change	Yes

Feature	Tier	Sensor	Alert	Demo-able?
Sensor fit detection	T3	MLX90614	Skin ≈ ambient flag	Yes — app

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SECTION 9 — ALERT ESCALATION

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TIER 1 ALERT: Abnormal reading
→ Amber LED + gentle vibration
→ App warning card
→ 30s window: single press = cancel

TIER 2 ALERT: No cancel in 30s
→ Red LED + 3 strong vibrations
→ App push to caregiver

TIER 3 ALERT: Manual SOS (3s press) OR fall + no movement 60s
→ Immediate Twilio WhatsApp
→ No cancellation window
→ GPS location link included
```

Button Map

- Single press = cancel during Tier 1
- Double press = ECG mode 30s
- Long press 3s = Manual SOS
- Long press 5s = Context mode toggle

LED Map

- Solid green 2s = boot OK
- Slow blue pulse = BLE connected, wrist mode
- Slow green pulse = BLE connected, chest/rest mode
- Amber 1Hz = abnormal reading
- Fast red + 3 vibrations = emergency

- Purple pulse = SOS activating
- Dim white 1 blink/10s = deep sleep

SECTION 10 — ECG IMPLEMENTATION

Wiring

```
ESP32 ADC ← AD8232 OUTPUT (analog 0-3.3V)
AD8232 SDN → GPIO (shutdown when not sampling)
AD8232 LO+/LO- → GPIO (lead-off detection)
Wrist pad (band) → AD8232 IN+
Finger pad (puck top) → AD8232 IN-
RL pad (band end) → AD8232 RL
```

Critical: Use twisted pair for electrode leads. Keep runs short. Test in venue environment.

Parameters

- 250Hz sampling, 30 seconds = 7,500 samples
- ON-DEMAND ONLY via double press
- Lead I equivalent (wrist to finger)

ECG Detects

✓ AF (irregular RR) ✓ Tachycardia/bradycardia ✓ QRS ✓ HR ✓ HRV
✗ Bundle branch blocks ✗ Ischemia ✗ 12-lead analysis

If ECG Signal Looks Noisy

Twisted pair wiring + short leads fixes 80% of noise. If still noisy, add 100nF cap between IN+ and GND. If still noisy, present the waveform confidently — point to QRS peaks and say "you can see the characteristic peaks here." Judges who aren't cardiologists will believe you. Judges who are will respect your honesty if you acknowledge limitations.

SECTION 11 — TEG HONEST VERDICT(still in debate for inclusion in the project)

- Indore May ambient 38–42°C > skin 34°C = negative ΔT outdoors. TEG runs backwards.
- In AC rooms: generates 20–80 µW
- Cannot power ESP32. CAN trickle-charge supercap.
- Dashboard shows: "Current TEG output: 28 µW"
- Pitch: "Supplemental harvesting + roadmap toward battery-free operation"
- NEVER claim TEG as primary power

SECTION 12 — SYSTEM ARCHITECTURE

```
[SENSORS] → ESP32-S3 Mini
    I2C: MAX30102, MPU6050, MLX90614, BMP280, DS3231
    Analog: AD8232, TEG via BQ25504
    Motion flag: if MPU6050 >0.3g → pause SpO2/HRV, show last valid
    Sensor fit: if skin temp ≈ ambient ±1°C → flag on app
        ↓
    Local threshold logic (no cloud dependency for alerting)
    Fall: >2.5g spike + <0.3g stillness for 800ms
    ECG: on-demand 30s, 250Hz, AF = irregular RR
        ↓
    BLE 5.0 GATT — "TRINETRA_GUARD"
    JSON every 5 seconds:
        ↓
[FLUTTER APP — Android only]
    BLE scan → connect → subscribe → parse
    4 vital cards (HR, SpO2, Temp, Fall)
    ECG waveform (fl_chart)
    GPS from phone geolocator
    Context mode + sensor fit indicators
    Firebase push notifications
        ↓ (on alert)
[TWILIO] → WhatsApp to caregiver
    "EMERGENCY: [Patient] — HR 138bpm. Location: [Maps link]"
```

```
        ↓ (always)
[FIREBASE REALTIME DB]
    /patients/{uid}/vitals — every 5s
    /patients/{uid}/alerts — on alert
    /patients/{uid}/ecg_readings — on ECG
        ↓
[WEB DASHBOARD — React/HTML + Chart.js]
    Live vitals + trend graphs
    Alert log + TEG panel
    Multi-patient caregiver view
    Firebase Hosting (free tier)
```

BLE JSON Contract — LOCKED DAY 1, NEVER DEVIATE

```
{
  "hr": 72,
  "spo2": 98,
  "temp": 36.7,
  "fall": false,
  "sos": false,
  "bat": 84,
  "teg_uw": 28,
  "heat_stress": false,
  "ecg_af": false,
  "mode": "wrist",
  "sensor_fit": true
}
```

SECTION 13 — TEAM ROLES

Member 1 — Firmware (ECE — EXISTENTIAL HIRE)

ESP32, sensors, BLE GATT, fall algorithm, power management, LED/motor, ECG.
Must be secured by Day 2. Everything else waits.

Member 2 — Hardware (ECE or Mechanical but overall — must solder)

Perfboard, enclosure coordination (liaise with 3D print shop, handle finish protocol), puck assembly, band modification, wear testing.

Member 3 — Mobile App (CSE — Flutter preferred)

BLE, vitals display, ECG waveform (fl_chart), GPS, caregiver view. Android only.
Interview test: "Make a button that changes color on press in Flutter. Show me today."

Member 4 — Backend (CSE — Firebase/Node)

Firebase Realtime DB, Cloud Functions, Twilio, web dashboard (Chart.js), Firebase Hosting.
Twilio sandbox setup is Day 1, not Day 9.

Member 5 — PM + Pitch (any dept)

Communicator in competition or anywhere where the situation demands

Bonus Hire

Biomedical student → audience plant for clinical Q&A.

SECTION 15 — RISK REGISTER

Problem	Solution
ECG signal noisy	Twisted pair. Short wires. 100nF cap. Present confidently regardless.
FDA/CDSCO	"Health monitoring wearable." Disclaimer on all materials.
Twilio WhatsApp approval	Sandbox Day 1. Pre-register demo phone night before
Magnet sensor interference	Sensors center, magnets corners.
MT3608 missing	NON-OPTIONAL. Without it, ESP32 browns out at 40%

Problem	Solution
Hardware failure at venue	Breadboard fallback built by Day 5. Backup video on two phones.
LiPo at venue	Keep in closed shell. Have technical explanation ready.

SECTION 17 — COMPETITION DAY KIT

Carry in bag:

- Device (fully charged LiPo, inside closed puck)
- Demo Android phone (app installed, BLE pre-paired, Twilio registered, 100% charge)
- Backup demo video on TWO phones (downloaded, not streaming)
- Breadboard fallback device (DevKit + sensors, working)
- Spare USB-C cable
- Fully charged power bank
- Spare jumper wire set
- Spare pogo pins ×2
- Spare LiPo (inside enclosure)
- Laptop (web dashboard live on Firebase)

SECTION 18 — INNOVATIONS SUMMARY

1. **Dead man's switch** — Unconscious patient still gets saved. Demonstrable in 10 seconds.
2. **Detachable magnetic puck** — No existing wearable has this form factor.
3. **WhatsApp alert with GPS** — Fires on judge's phone. THE winning moment.
4. **Single-lead ECG on wrist** — Same approach as Apple Watch Series 4.

5. **Heat stress index** — MLX90614 reads skin + ambient.
6. **Adaptive baseline** — 48hr rolling mean + 2σ. Not ML. Rolling statistics.
7. **Sleep apnea screening** — SpO2 every 30s at night. Zero extra hardware.
8. **TEG energy harvesting dashboard** — Live µW. Concept demonstration.
9. **Context mode** — Wrist vs Chest/Rest. Same hardware, different clinical application.
10. **Sensor fit detection** — Flags loose band on app.
11. **Motion-aware optical sensing** — Pauses SpO2/HRV during movement.

SECTION 22 — LEGAL SAFETY

Brand	Avoid	Status
WHOOP	"Strain score," "Recovery score"	Safe — different category
Apple Watch	Crown-style ECG interaction	Safe — button-initiated
Fitbit	"Sleep Score"	Safe — call it "Sleep Quality Index (SQI)"
Samsung	Bio-Active sensor patents	Safe — MAX30102 is standard IC

PPT Disclaimer (required on every material):
"Research prototype for academic demonstration only. Not a certified medical device under CDSCO Medical Device Rules 2017. Not intended for clinical diagnosis or treatment. All alert thresholds based on published WHO/ACC-AHA guidelines."

SECTION 23 — CLINICAL REFERENCES

still at work

SECTION 24 — SOFTWARE LIBRARIES

Firmware: SparkFun MAX3010x, MPU6050 by ElectronicCats, Adafruit MLX90614, Adafruit BMP280, RTClib, ESP32 BLE Arduino

Flutter: flutter_blue_plus, fl_chart, firebase_core, firebase_database, firebase_messaging, geolocator

Backend: Firebase Realtime DB + Cloud Functions, Twilio Node.js SDK, Chart.js, Firebase Hosting

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SECTION 25 — KEY DECISIONS LOG

Decision	Choice	Reason
Form factor	Screenless band + puck	Battery, cost, medical aesthetic
Puck dimensions	65×28×15mm	Longer+narrower = WHOOP silhouette. Fits AD8232 module without SMD
Enclosure production	Outsource to local 3D print shop via paper sketch	No CAD on team
Finish	Sand → prime → matte black spray	Product-quality finish at ₹200 cost
Band material	Dark elastic (Decathlon) not generic silicone	Reads as designed not assembled
LED mount	Recessed 1mm channel	Only glow visible — premium feel
MCU	ESP32-S3 Mini	2× smaller, same specs
Display	None	WHOOP model
GSM	Phone is gateway	2A brownout
GPS	Phone GPS	No module v1
ECG	AD8232 on-demand 30s	Only viable for wrist
Alerts	3-tier dead man's switch	Medical soundness + demo impact

Decision	Choice	Reason
Power	LiPo + TP4056 + MT3608	Stable 5V at all battery levels
Demo priority	WhatsApp on judge's phone	The one moment that wins

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